

space complexity of
bfs and dfs

CSCI
270

what we know so far

runtime

bfs $O(b^d)$

dfs $O(b^m)$

solution depth

d

the length of the shortest search path that ends at a final state

maximum depth

m

the length of the longest search path

branching factor

b

the maximum number of successors of a state

but what about this?

runtime

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space can be measured
by the **maximum number**
of nodes in the **container**
at any given time



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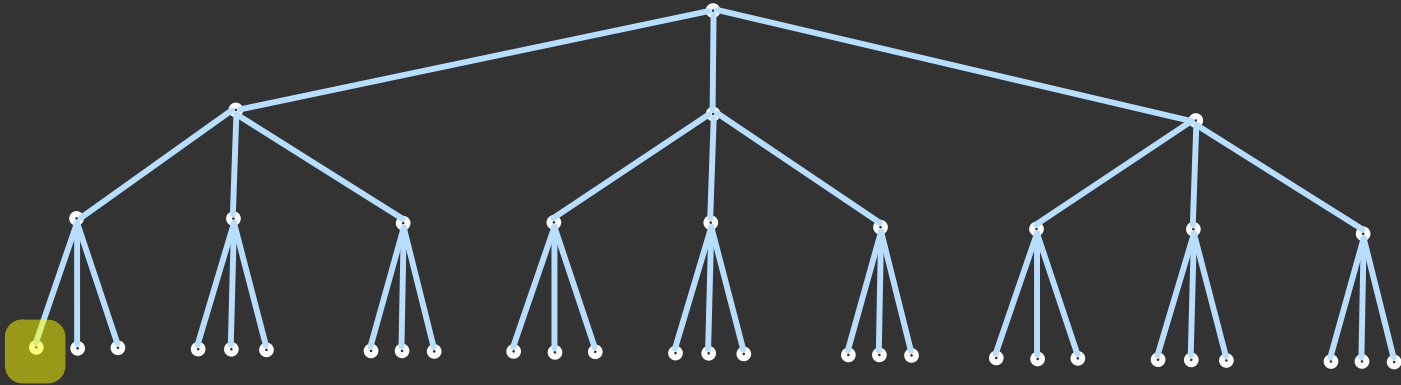
the length of the
longest search path

branching factor

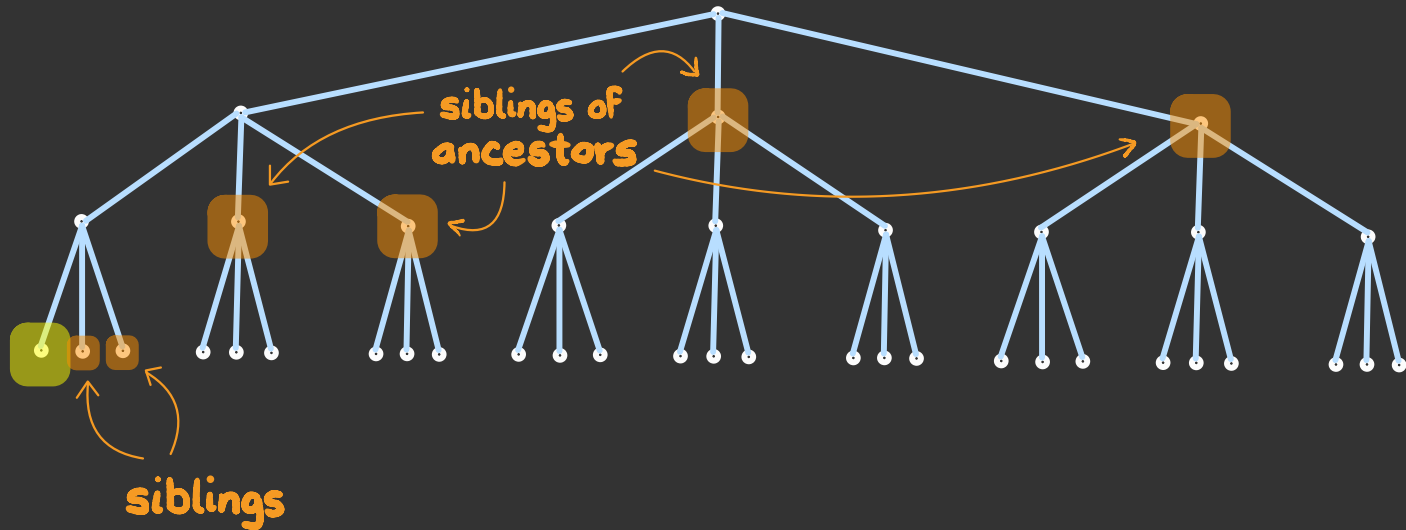
b

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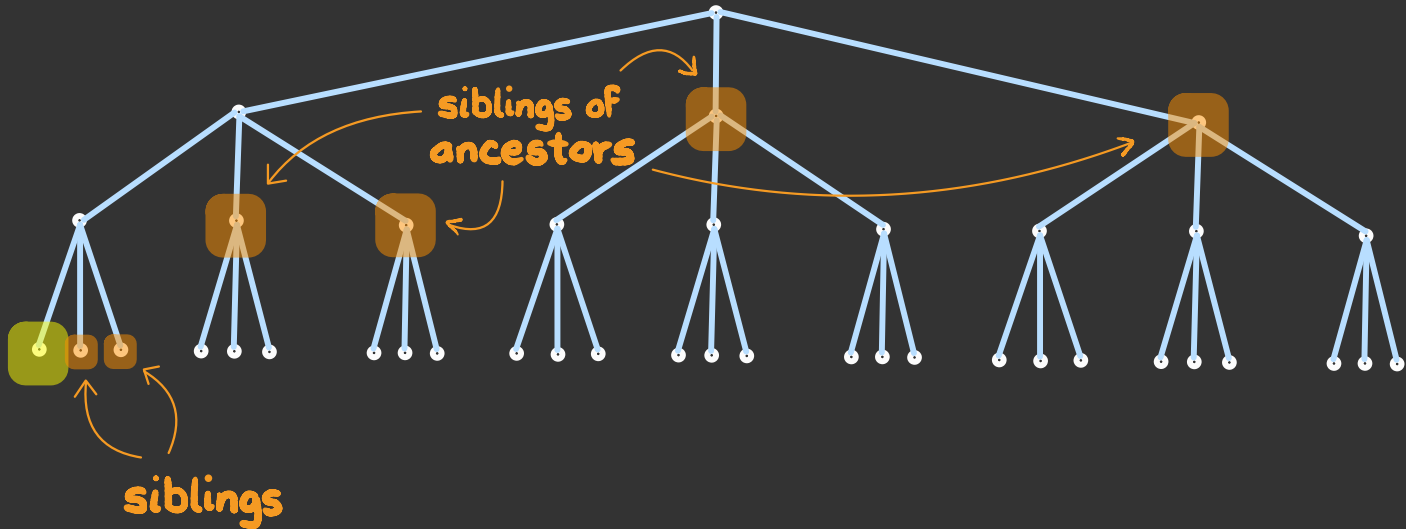
for **dfs**, what search nodes might be in the container when we visit the **highlighted node**?



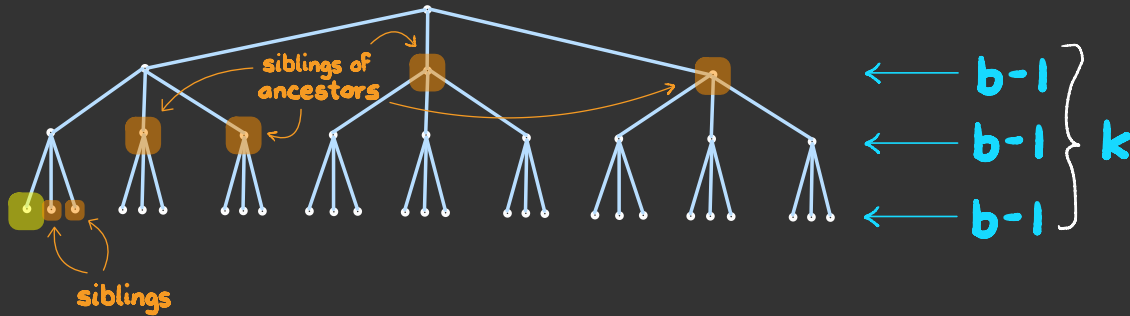
for **dfs**, what search nodes might be in the container when we visit the **highlighted node**?



if the **highlighted node** is at depth **k**,
how many nodes is this?

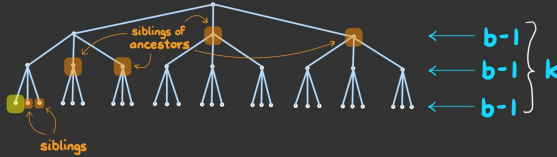


if the highlighted node is at depth k ,
how many nodes is this?



$$k(b-1) = 0(bk)$$

if the **highlighted node** is at depth **k**,
how many nodes is this?



$$k(b-1) = O(bk)$$

given this observation, what
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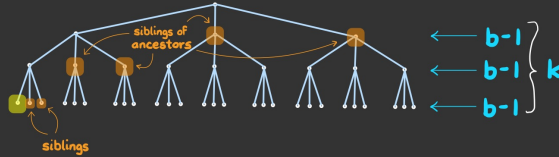
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$$O(bm)$$

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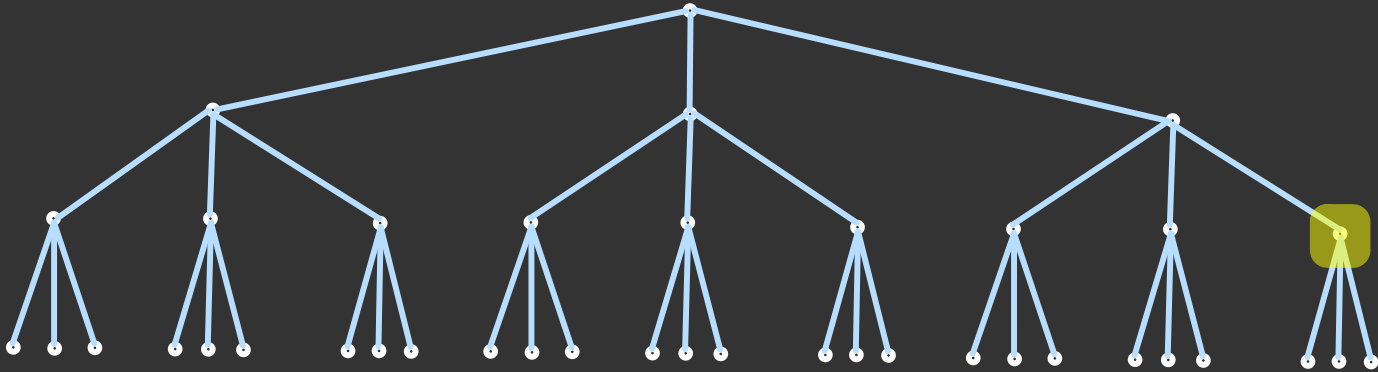
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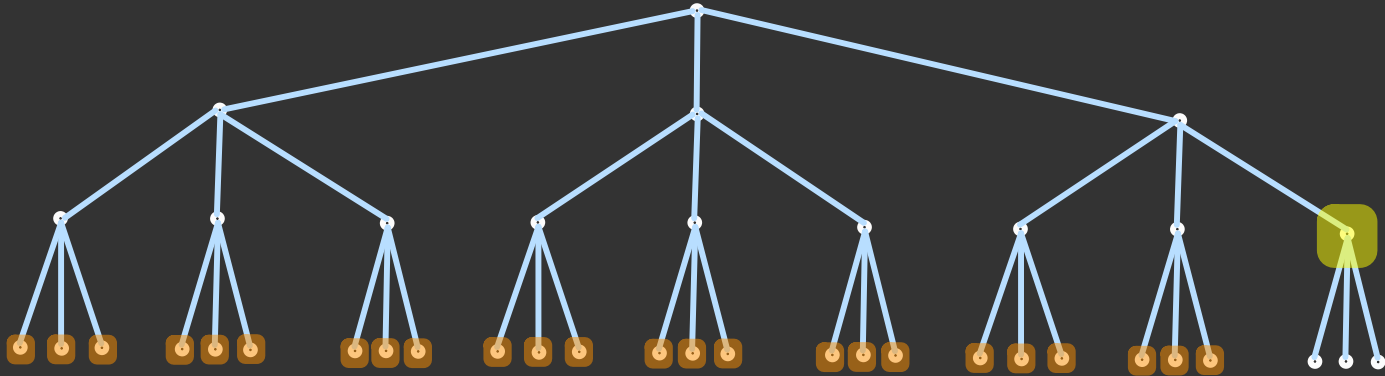
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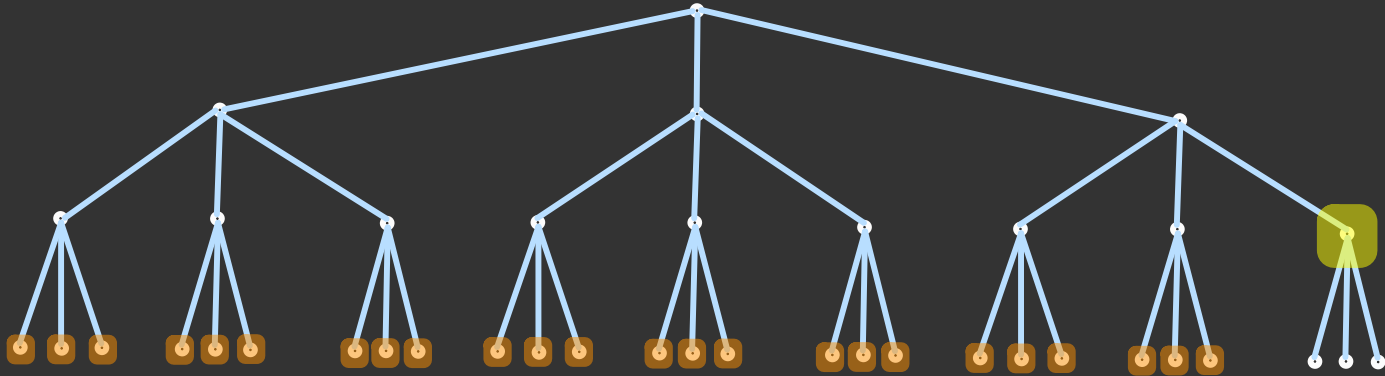
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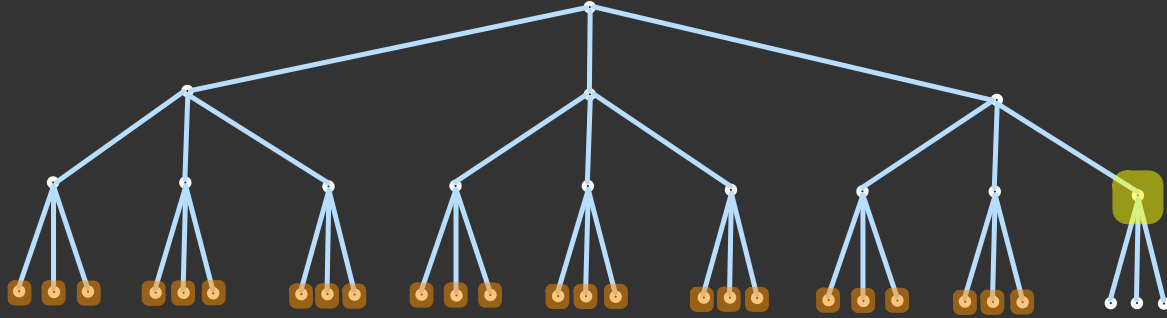
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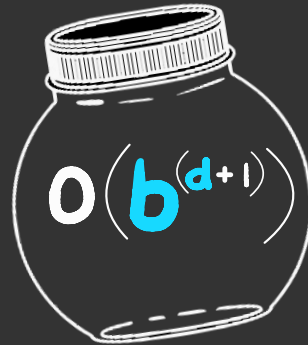
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which typically matters more?

runtime  space

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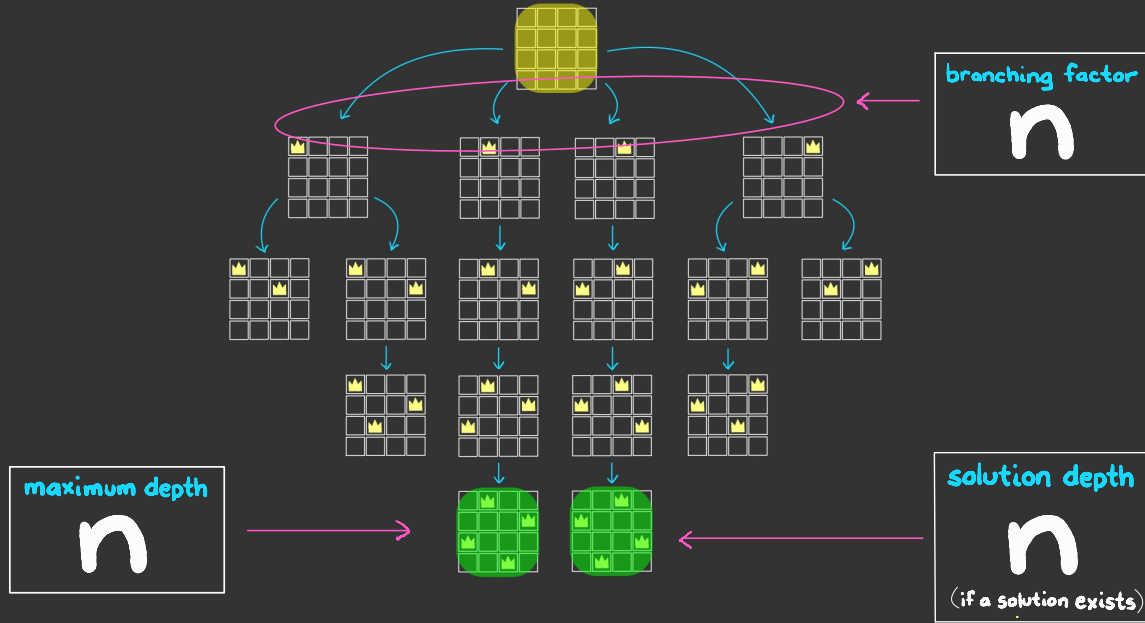
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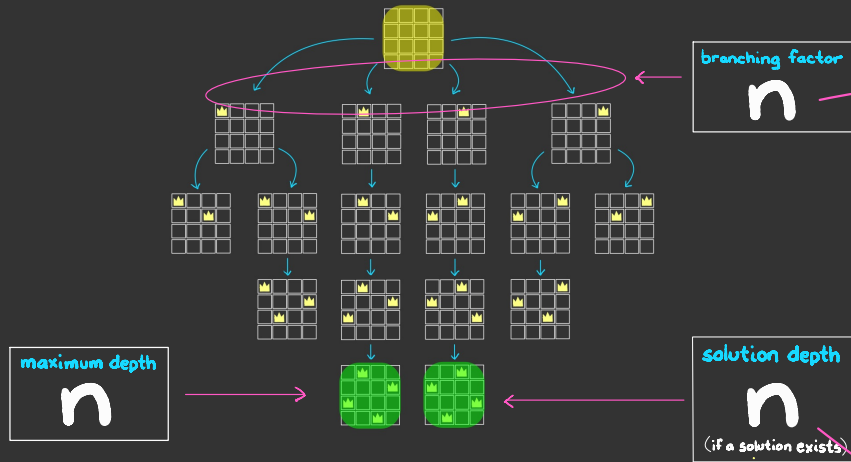
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why space?

recall this formulation of n-queens

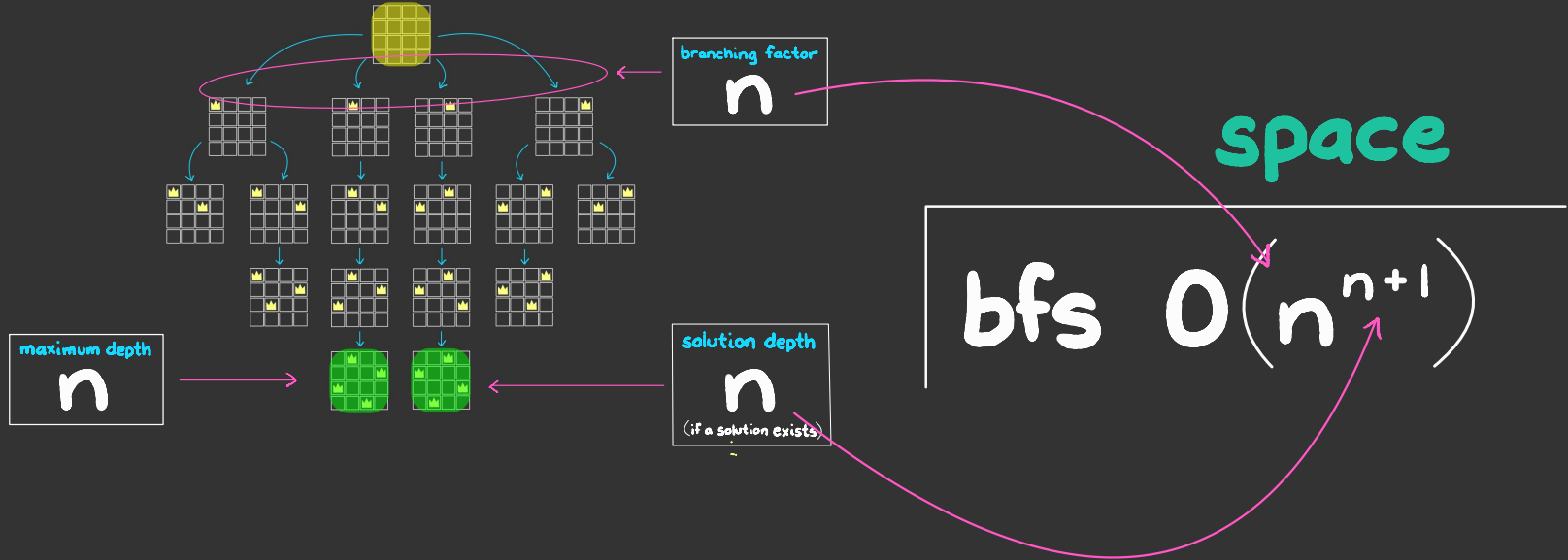




space

$$\text{bfs } O(b^{(d+1)})$$

even for the 10-queens problem, we
require $10^{10} \approx 100\text{GB}$ of memory
(assuming we can cram each search node into a byte)



this **seriously** limits the applicability of bfs in practice

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